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Antibiotic susceptibility prediction using transformers

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Background

- Antibiotic treatment depends on the susceptibility of the bacteria, typically estimated using cultivation-based methods that can be slow and time-consuming.
- Physicians' empirical treatment has a considerable chance of failing or being unnecessarily broad.

Contribution

- Application of Natural Language Processing tools to predict antibiotic resistance from incomplete data.
- Development of a deep learning model that predicts the susceptibility of bacterial isolates to antibiotics.
- Uncertainty control to make predictions with pre-specified confidence for all possible outputs.

Dataset

- Susceptibility test data for 413,593 isolates from 30 European countries retrieved from The European Centre for Disease Prevention and Control.
- The isolates correspond to *Escherichia coli* obtained from blood samples.

Model

- Transformer encoder + artificial neural networks.
- Conditional inductive conformal prediction for uncertainty control.
- The softmax outputs of the neural networks indicate the probability of susceptibility or resistance of the isolate to that antibiotic.
- The model is train by minimizing the cross-entropy loss over all predicted antibiotics.

Input

- Partial diagnostic information and patient data (gender, age, country, date).
- The patient information and the partial antibiotic susceptibility data are preceded by a CLS token and padded.

Output

- A complete susceptibility profile for 16 antibiotics based on the softmax outputs of the model.
- A prediction region: {}, {S}, {R}, or {S, R} for each antibiotic containing the true susceptibility profile at a fixed confidence level.
- The prediction regions are built based on the deviation of the conformity scores from a calibration distribution.

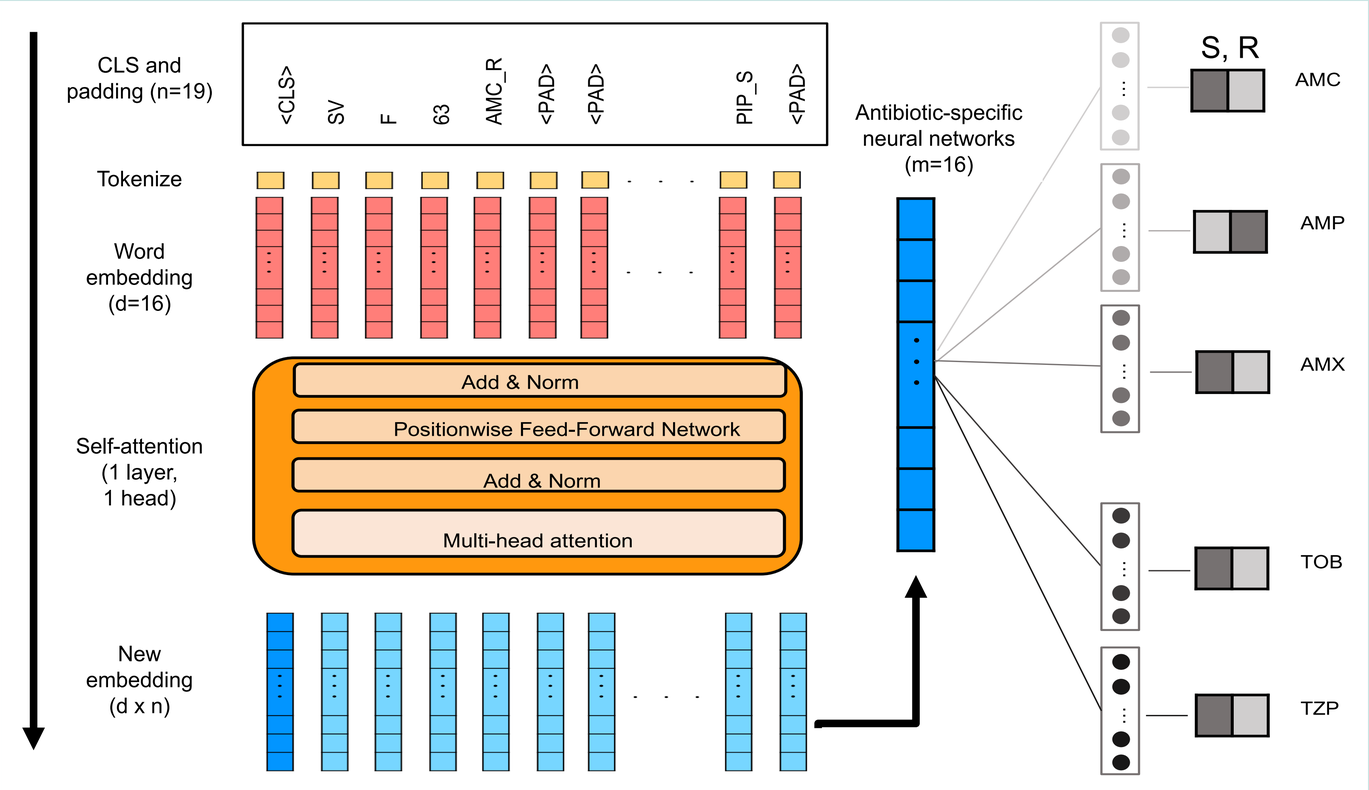

Figure 1. The architecture of the model.

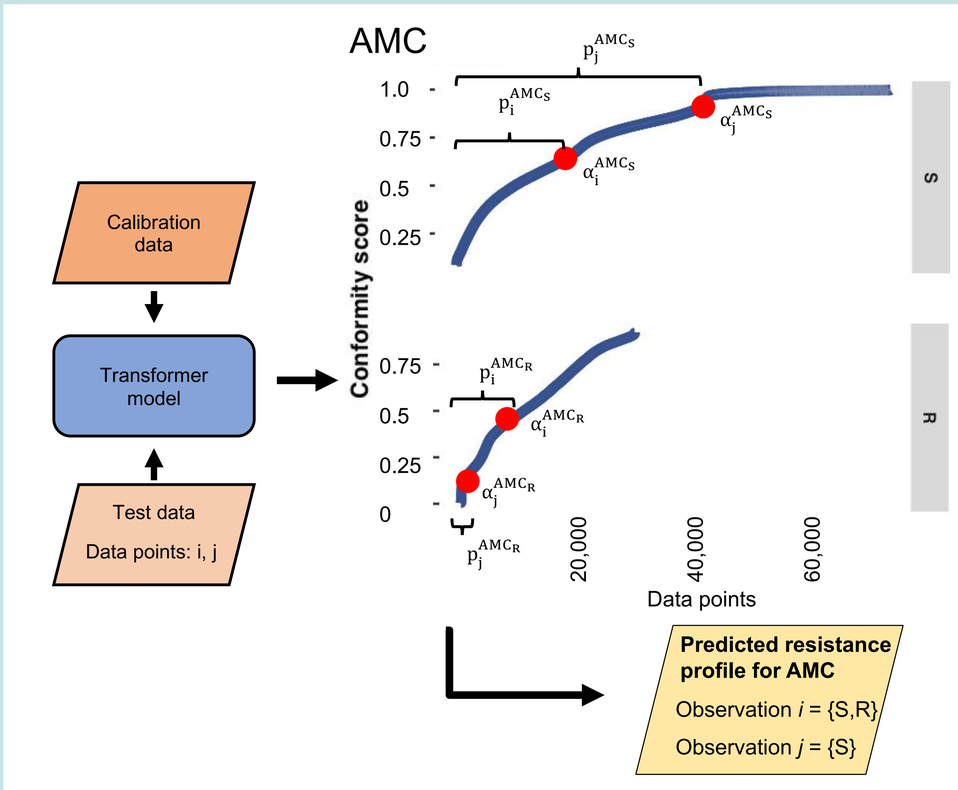

Figure 2. Uncertainty control.

Results

- Susceptibility is predicted with an error rate below 5% on average for quinolones and cephalosporins, and below 15% for penicillins and aminoglycosides (major error).
- Resistance to cephalosporins, quinolones and aminoglycosides is predicted with an average error rate between 10% and 20%, and 24% for penicillins (very major error).
- The uncertainty control algorithm allows us to fix the empirical error rate for both susceptible and resistant isolates to a pre-specified level.

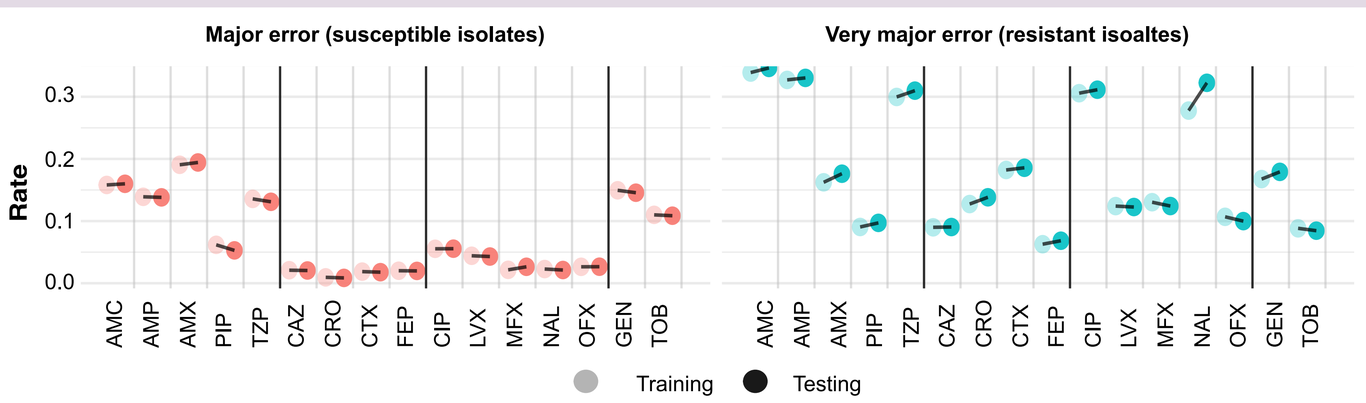

Figure 3. Major error and very major error results from the cross-validation.

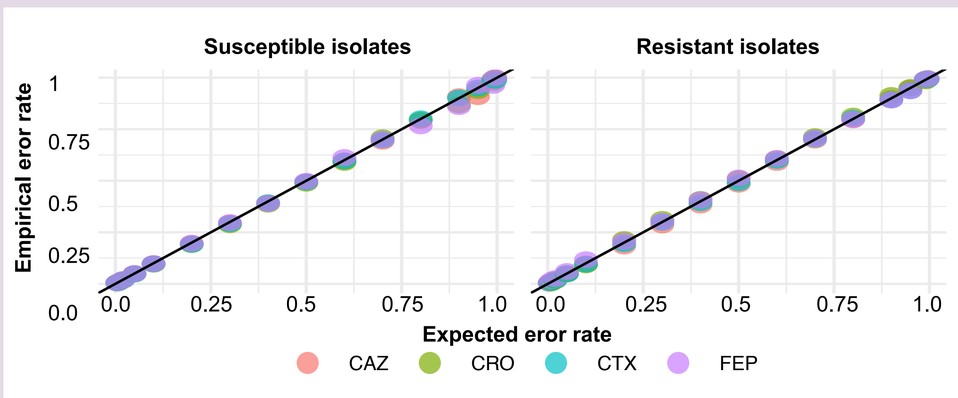

Figure 5. Empirical and expected error rates for cephalosporins after uncertainty control.

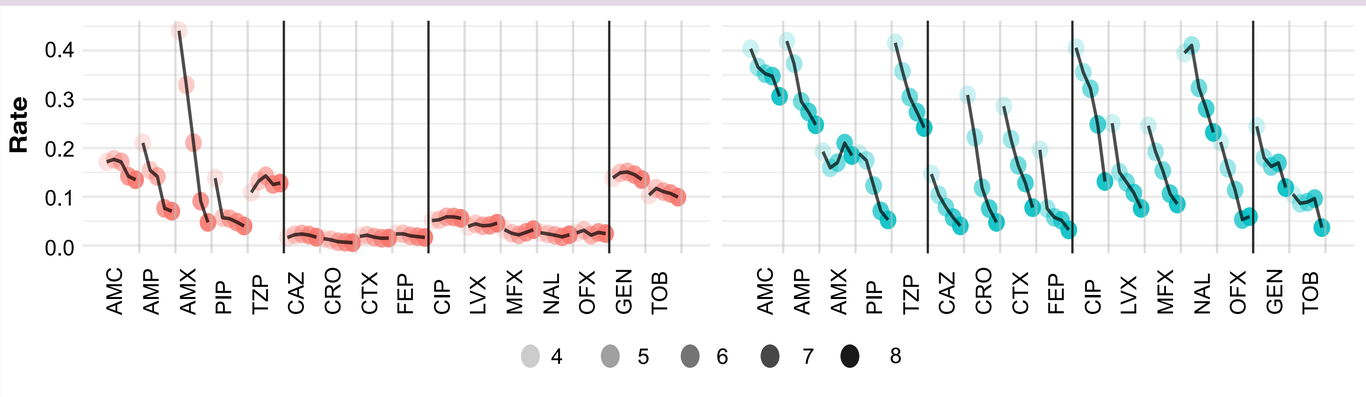

Figure 4. The predictive performance of the model as a function of the number of tested antibiotics included in the input.

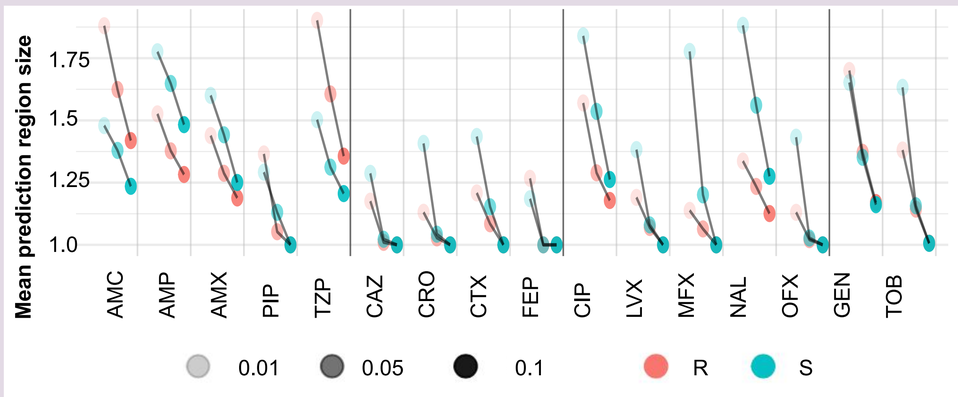

Figure 6. The average of the prediction region size as a function of the confidence level for resistant (R) and susceptible (S) isolates.

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